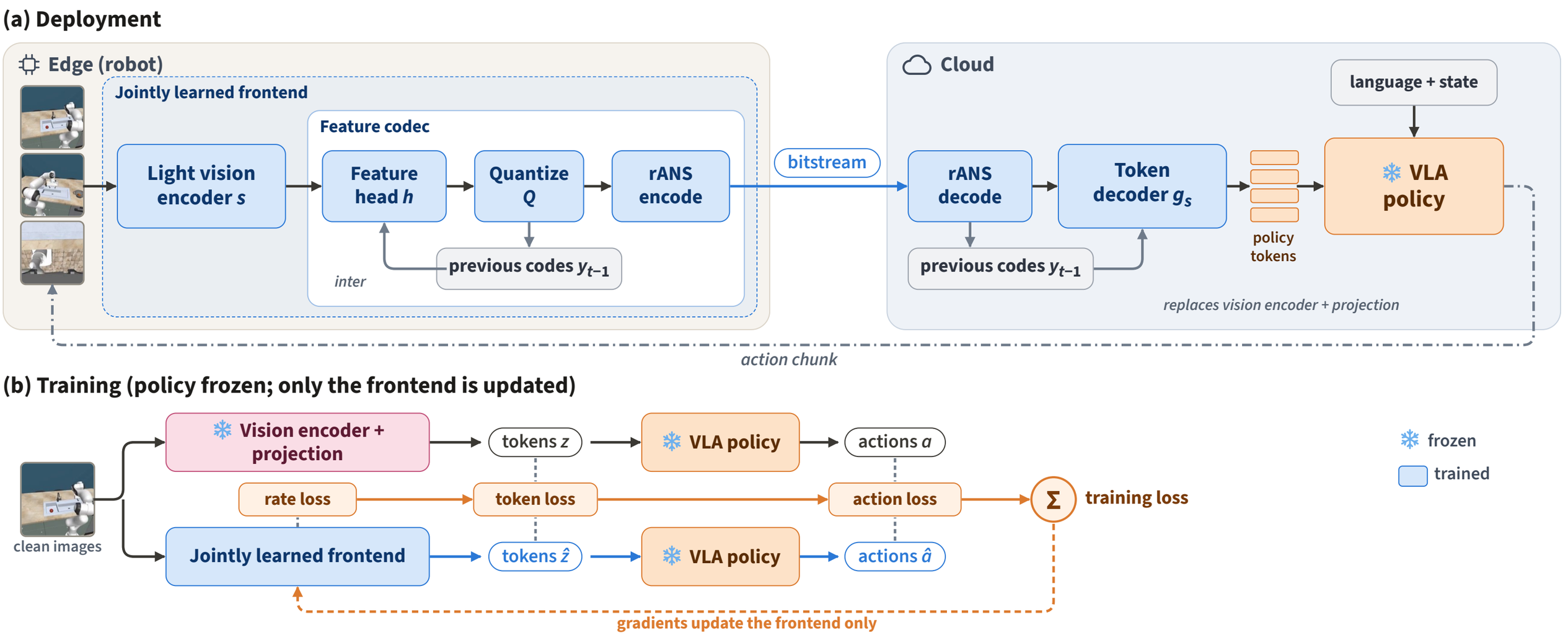

(a) Deployment
Edge (robot)
Jointly learned frontend
Feature codec
Light vision encoder s
Feature head h
Quantize Q
rANS encode
previous codes y_{t-1}
inter
bitstream
Cloud
language + state
rANS decode
Token decoder g_s
policy tokens
VLA policy
previous codes y_{t-1}
replaces vision encoder + projection
action chunk
(b) Training (policy frozen; only the frontend is updated)
clean images
Vision encoder + projection
tokens z
VLA policy
actions a
rate loss
token loss
action loss
training loss
Jointly learned frontend
tokens ẑ
VLA policy
actions â
gradients update the frontend only
frozen
trained